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Kernel Density Estimation with Polyspherical Data and its Applications

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ABSTRACT

A kernel density estimator for data on the polysphere $\mathbb{S}^{d_1} \times \dots \times \mathbb{S}^{d_r}$, with $r, d_1, \dots, d_r \geq 1$, is presented in this article. We derive the main asymptotic properties of the estimator, including mean square error, normality, and optimal bandwidths. We address the kernel theory of the estimator beyond the von Mises–Fisher kernel, introducing new kernels that are more efficient and investigating normalizing constants, moments, and sampling methods thereof. Plug-in and cross-validated bandwidth selectors are also obtained. As a spin-off of the kernel density estimator, we propose a nonparametric k -sample test based on the Jensen–Shannon divergence that is consistent against alternatives with non-homogeneous densities. Numerical experiments illuminate the asymptotic theory of the kernel density estimator and demonstrate the superior performance of the k -sample test with respect to parametric alternatives in certain scenarios. Our smoothing methodology is applied to the analysis of the morphology of a sample of hippocampi of infants embedded on the high-dimensional polysphere $(\mathbb{S}^2)^{168}$ through skeletal representations (s -reps). Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

Directional data are observations recorded as directions or angles that are typically represented on the circumference, the sphere, or the Cartesian products thereof with other manifolds. In the spirit of Eltzner, Jung, and Huckemann (2015), we use the term *polyspherical data* to refer to observations on the product space of spheres $\mathbb{S}^{d_1} \times \dots \times \mathbb{S}^{d_r}$, where $d_1, \dots, d_r \geq 1$ and $\mathbb{S}^d$ denotes the (hyper)sphere of dimension $d \geq 1$. The polysphere encompasses particular important cases, such as the circle ($r = d_1 = 1$), the sphere ($r = 1, d_1 = 2$), and the torus ($d_1 = \dots = d_r = 1$). Specific instances of polyspherical data include medically imaged objects (e.g., Vuollo et al. 2016; Pizer et al. 2022), protein/RNA structures (e.g., Boomsma et al. 2008; Eltzner, Huckemann, and Mardia 2018), and several studies within the field of astronomy, such as cosmic rays and crater locations (e.g., Baldi et al. 2009; García-Portugués, Navarro-Esteban, and Cuesta-Albertos 2023), among others. More generally, high-dimensional spherical data appear when standardizing multivariate data by Euclidean norms, as done in genetics (e.g., Eisen et al. 1998) and text mining (e.g., Banerjee et al. 2005), when considering *pres*shapes in shape analysis (Dryden and Mardia 2016), or when applying embeddings of different data types to the sphere (e.g., Li, Yoon, and Ahn 2023; Zouboulouglou, García-Portugués, and Marron 2023). Mardia and Jupp (1999) and Ley and Verdebout (2017) collect fundamental methods on directional statistics, while Pewsey and García-Portugués (2021) provides a survey of recent advances in this field. In addition, Marron and Dryden (2021)

reviews novel methodology and different case studies involving directional data.

In the context of three-dimensional shapes, effective methods for parameterization of their structures are skeletal representations, referred to as s -reps (Pizer et al. 2013). Skeletal models consist of a centrally located skeleton, and spokes represented as vectors that emanate from the skeletal structure and terminate at the boundary of the object (see Figure 1(a)). Consequently, these models provide benefits compared to those that solely capture its boundary (see, e.g., Liu et al. 2021). The geometry characteristics of s -reps, particularly the directions represented by the spokes, imply that they reside within a space involving a Cartesian product of spheres. Skeletal models have been particularly used to analyze three-dimensional shapes derived from medical imaging. For instance, Pizer et al. (2013) provided a method for analyzing the shape of the hippocampus using nested sphere statistics of skeletal models, revealing a difference in hippocampus shapes between infants diagnosed as autistic and non-autistic. Schulz et al. (2016) introduced a permutation test to identify morphological differences between two populations based on geometric features extracted from their s -reps. The approach uses Composite Principal Nested Spheres (CPNS), a composition of Principal Nested Spheres (PNS; Jung, Dryden, and Marron 2012) with Principal Component Analysis (PCA) to detect the main source of variation in the space of s -reps. More recently, Liu et al. (2023b) conducted multi-object shape analysis to extract joint shape variation between two functionally related brain structures, the hippocampus and the caudate. Liu et al. (2023a) demonstrated the effectiveness

of jointly analyzing within- and between-object shape features, resulting in improved classification and hypothesis testing performance. From a different perspective, García-Portugués and Meilán-Vila (2023) identified the primary source of variability in hippocampus shapes using nonparametric density ridge estimation. For further applications involving hippocampus shapes parameterized using s -reps, we refer to the recent survey by Pizer et al. (2022).

This work tackles the problem of nonparametric density estimation with polyspherical data. In the statistical literature, considerable research has been done on nonparametric density estimation within directional supports, mostly based on kernel smoothing approaches (Pewsey and García-Portugués 2021, sec. 7.1). The seminal works by Hall, Watson, and Cabrera (1987) and Bai, Rao, and Zhao (1988) defined three highly related kernel density estimators on $\mathbb{S}^d$, $d \geq 1$, deriving their asymptotic formulae of bias and variance (Hall, Watson, and Cabrera 1987), and establishing its strong uniform and L^1 -norm consistency (Bai, Rao, and Zhao 1988). Somehow more explicit derivations of the mean-square error analysis, for Bai, Rao, and Zhao's (1988) definition, were later given in Zhao and Wu (2001) and García-Portugués, Crujeiras, and González-Manteiga (2013). Klemelä (2000) presented an alternative estimator for the density and its derivatives on $\mathbb{S}^d$, $d \geq 2$, using a kernel featuring an intrinsic distance. Di Marzio, Panzera, and Taylor (2011) introduced a specific class of product kernels to construct a kernel density estimator on the torus, while García-Portugués, Crujeiras, and González-Manteiga (2013) constructed an estimator on $\mathbb{S}^d \times \mathbb{R}$, also using product kernels. In a broader context, the problem of nonparametric density estimation on Riemannian manifolds has been investigated by Pelletier (2005), Henry and Rodriguez (2009), and Wu and Wu (2022), among others. Parallel to the development of these density estimators, several bandwidth selection methods have been developed for them. Taylor (2008) introduced a plug-in bandwidth selector for circular data, while Oliveira, Crujeiras, and Rodríguez-Casal (2012) derived a mixture-based selector. Di Marzio, Panzera, and Taylor (2011) proposed a cross-validatory bandwidth selection method for the torus. For $\mathbb{S}^d$, Hall, Watson, and Cabrera (1987) provided two cross-validatory bandwidth selectors, and García-Portugués (2013) gave a rule-of-thumb bandwidth selector, assuming the underlying density is von Mises–Fisher. Kernel density estimators on directional supports have also been used to guide nonparametric statistical inference, such as in goodness-of-fit (Boente, Rodriguez, and González-Manteiga 2014; García-Portugués, Crujeiras, and González-Manteiga 2015), mode (Ameijeiras-Alonso et al. 2019), or independence testing (García-Portugués, Crujeiras, and González-Manteiga 2015).

In this article, we propose a kernel density estimator for data on the polysphere $\mathbb{S}^{d_1} \times \dots \times \mathbb{S}^{d_r}$. The estimator accommodates previous estimators as particular cases and has manifold associated methodological contributions. The first set of contributions involves the main asymptotic properties of the estimator, namely, mean-square error analysis, consistency, normality, and optimal bandwidths. Unlike previous specific approaches, we highlight that these results are obtained for general non-product kernels, and that the asymptotic bias is derived with

a smaller remaining order. These properties are exemplified through a numerical study that enlightens the role of the bandwidth sequence on the asymptotic normality. The second series of contributions relates to the definition and efficiency analysis of other kernels beyond the von Mises–Fisher kernel, which is omnipresent in kernel density estimation with directional data. In this regard, we introduce several more efficient kernels for both the $r = 1$ and $r > 1$ cases, addressing practically relevant matters such as normalizing constants, moments, and simulation methods. The third contribution consists of deriving an inexpensive rule-of-thumb plug-in bandwidth selector and formulating appropriate cross-validation selectors. The fourth contribution entails the introduction of a new k -sample permutation test based on the kernel density estimator and Jensen–Shannon divergence that, unlike previous smoothing tests, is applicable in a high-dimensional space. The test is seen to be consistent against all alternatives with non-homogeneous densities. Numerical experiments show a better performance of this k -sample test compared to parametric alternatives in certain scenarios, as well as its applicability on high dimensions. Fifth, our methodology is applied to analyze the hippocampus shapes of 6-months-old infants (Liu et al. 2021, 2023a) that are embedded on the high-dimensional polysphere $(\mathbb{S}^2)^{168}$ through s -reps. In this application, we identify the most prototypical and outlying hippocampi, as well as find a mild significance of the shape difference between children who were later diagnosed as autistic and non-autistic.

The rest of this article is organized as follows. In Section 2, the nonparametric estimator of the density function is introduced. Section 3 provides asymptotic results on the bias, variance, optimal bandwidths, and normality of the estimator. Section 4 is devoted to kernel theory, addressing the derivation of new kernels, their efficiency, and other useful results. Section 5 gives bandwidth selectors based on plug-in and cross-validation approaches. Section 6 introduces a novel k -sample test based on the kernel density estimator. Section 7 presents two applications of the density estimator to the analysis of hippocampus shapes. The article concludes with a discussion in Section 8. Proofs and numerical experiments are relegated to the supplementary material (SM). The R package `polykde` (<https://CRAN.R-project.org/package=polykde>) implements the presented methods and allows replicating the empirical results of the article.

2. Estimator

Let f be a probability density function (pdf) on $\mathbb{S}^{\mathbf{d}} := \mathbb{S}^{d_1} \times \dots \times \mathbb{S}^{d_r}$, $\mathbf{d} = (d_1, \dots, d_r)'$ and $d_1, \dots, d_r \geq 1$, with respect to the product measure $\sigma_{\mathbf{d}} := \sigma_{d_1} \times \dots \times \sigma_{d_r}$, where σ_{d_j} is the surface area measure on $\mathbb{S}^{d_j}$. We denote $\omega_{d_j} := \sigma_{d_j}(\mathbb{S}^{d_j}) = 2\pi^{(d_j+1)/2} / \Gamma((d_j+1)/2)$ and $\omega_{\mathbf{d}} := \prod_{j=1}^r \omega_{d_j}$ to the surface areas of $\mathbb{S}^{d_j}$ and $\mathbb{S}^{\mathbf{d}}$, respectively. We denote $\tilde{\mathbf{d}} := \sum_{j=1}^r d_j$, and so $\mathbb{S}^{\mathbf{d}} \subset \mathbb{R}^{\tilde{\mathbf{d}}+r}$.

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be an iid sample from f . Let $\mathbf{x} = (\mathbf{x}'_1, \dots, \mathbf{x}'_r) \in \mathbb{S}^{\mathbf{d}}$, with $\mathbf{x}_j \in \mathbb{S}^{d_j}$ for $j = 1, \dots, r$, and set $\mathbf{h} := (h_1, \dots, h_r)' \in \mathbb{R}_+^r$. Building on Bai, Rao, and Zhao's (1988) definition, our kernel density estimator (kde) of f at $\mathbf{x}$ is defined as

$$\hat{f}(\mathbf{x}; \mathbf{h}) := \frac{1}{n} \sum_{i=1}^n L_{\mathbf{h}}(\mathbf{x}, \mathbf{X}_i), \quad (1)$$

$$L_{\mathbf{h}}(\mathbf{x}, \mathbf{y}) := c_{d,L}(\mathbf{h}) L\left(\frac{1 - \mathbf{x}'_1 \mathbf{y}_1}{h_1^2}, \dots, \frac{1 - \mathbf{x}'_r \mathbf{y}_r}{h_r^2}\right),$$

where the normalized kernel $L_{\mathbf{h}} : \mathbb{S}^d \times \mathbb{S}^d \rightarrow \mathbb{R}_{\geq 0}$ is based on the kernel $L : \mathbb{R}_{\geq 0}^r \rightarrow \mathbb{R}_{\geq 0}$. Applying r marginal tangent-normal changes of variables (see, e.g., Lemma 2 in García-Portugués, Crujeiras, and González-Manteiga 2013), the normalizing constant of L is

$$\begin{aligned} c_{d,L}(\mathbf{h})^{-1} &:= \int_{\mathbb{S}^d} L\left(\frac{1 - \mathbf{x}'_1 \mathbf{y}_1}{h_1^2}, \dots, \frac{1 - \mathbf{x}'_r \mathbf{y}_r}{h_r^2}\right) \sigma_d(d\mathbf{x}) \\ &= \int_{[-1,1]^r} L\left(\frac{1 - t_1}{h_1^2}, \dots, \frac{1 - t_r}{h_r^2}\right) \\ &\quad \times \prod_{j=1}^r (1 - t_j^2)^{d_j/2-1} \omega_{d_j-1} dt \\ &= \rho(\mathbf{h}) \lambda_{\mathbf{h},d}(L) \asymp \rho(\mathbf{h}) \lambda_d(L), \end{aligned} \quad (2)$$

with $a_\ell \asymp b_\ell$ denoting $a_\ell/b_\ell \rightarrow 1$ as $\ell \rightarrow \infty$ in (2), $I_{\mathbf{h}} := [0, 2h_1^{-2}] \times \dots \times [0, 2h_r^{-2}]$, $\lambda_{\mathbf{h},d}(L) := \omega_{d-1} \int_{I_{\mathbf{h}}} L(\mathbf{s}) \times \prod_{j=1}^r s_j^{d_j/2-1} (2 - s_j h_j^2)^{d_j/2-1} d\mathbf{s}$, and

$$\rho(\mathbf{h}) := \prod_{j=1}^r h_j^{d_j} \quad \text{and}$$

$$\lambda_d(L) := \omega_{d-1} \int_{\mathbb{R}_+^r} L(\mathbf{s}) \times \prod_{j=1}^r s_j^{d_j/2-1} 2^{d_j/2-1} d\mathbf{s}.$$

The most popular kernel on the sphere is the von Mises–Fisher (vMF) kernel, defined as $L_{\text{vMF}}(t) := e^{-t}$ for $t \geq 0$. It is related with the widely known vMF distribution on $\mathbb{S}^d$, $d \geq 1$, whose pdf is

$$\begin{aligned} f_{\text{vMF}}(\mathbf{x}; \boldsymbol{\mu}, \kappa) &:= c_d^{\text{vMF}}(\kappa) e^{\kappa \mathbf{x}' \boldsymbol{\mu}}, \\ c_d^{\text{vMF}}(\kappa) &:= \frac{\kappa^{(d-1)/2}}{(2\pi)^{(d+1)/2} \mathcal{I}_{(d-1)/2}(\kappa)}, \end{aligned}$$

where $\boldsymbol{\mu} \in \mathbb{S}^d$ is the spherical location, $\kappa \geq 0$ is the concentration parameter about $\boldsymbol{\mu}$, and $\mathcal{I}_\nu$ is the modified Bessel function of the first kind of order ν . For this kernel, it easily follows that $c_{d,L_{\text{vMF}}}(\mathbf{h})^{-1} = c_d^{\text{vMF}}(1/h^2)^{-1} e^{-1/h^2}$. If the product kernel $L_{\text{vMF}}^P(\mathbf{s}) := \prod_{j=1}^r L_{\text{vMF}}(s_j)$ is used in (1), the kde becomes a mixture of componentwise independent vMF pdfs:

$$\begin{aligned} \hat{f}(\mathbf{x}; \mathbf{h}) &= \frac{1}{n} \sum_{i=1}^n f_{\text{PvMF}}(\mathbf{x}; \mathbf{X}_i, \mathbf{h}^{\odot(-2)}), \\ f_{\text{PvMF}}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\kappa}) &:= \prod_{j=1}^r f_{\text{vMF}}(\mathbf{x}_j; \boldsymbol{\mu}_j, \kappa_j), \end{aligned} \quad (3)$$

where $\odot$ denotes the Hadamard product operator, PvMF means *Polyspherical vMF*, and $\mathbf{x}, \boldsymbol{\mu} \in \mathbb{S}^d$ and $\boldsymbol{\kappa} \in \mathbb{R}_+^r$. In this case, $c_{d,L_{\text{vMF}}}^P(\mathbf{h})^{-1} = c_d^{\text{vMF}}(\mathbf{h}^{\odot(-2)})^{-1} \prod_{j=1}^r e^{-1/h_j^2}$, where $c_d^{\text{vMF}}(\boldsymbol{\kappa}) = \prod_{j=1}^r c_{d_j}^{\text{vMF}}(\kappa_j)$.

Another relevant kernel is the *Epanechnikov* kernel $L_{\text{Epa}}(t) := (1-t)1_{\{0 \leq t \leq 1\}}$ (henceforth abbreviated as “Epa”) defined in Hall, Watson, and Cabrera (1987). In that paper, it was stated (p. 758) that this kernel is the most efficient for (1), when $r = 1$ and, at least, $d = 2$ (the statement allows for some degree of ambiguity about its scope in d). The precise quantification of this efficiency on $\mathbb{S}^d$ will be addressed in Section 4.

The estimator (1) accommodates as particular cases the usual kdes by Beran (1979, p. 1169), Bai, Rao, and Zhao (1988, p. 25), and Hall, Watson, and Cabrera (1987, p. 52) when $r = 1$ and $d_1 \geq 1$. When $d_1 = \dots = d_r = 1$, this kde is closely related to that proposed by Di Marzio, Panzera, and Taylor (2011, p. 2158) on the torus after a reparameterization. Specifically, set $\mathbf{h} = \boldsymbol{\kappa}^{\odot(-1/2)}$, where $\boldsymbol{\kappa}$ is the vector of the concentrations appearing in Di Marzio, Panzera, and Taylor (2011). Then, introducing polar coordinates, the kde (1) restricted to $d_1 = \dots = d_r = 1$ and the one presented by Di Marzio, Panzera, and Taylor (2011) coincide for a common vMF kernel. Within the toroidal setting, Di Marzio, Panzera, and Taylor (2011)’s kde is more general, as it also targets density derivative estimation, which we do not pursue in this article. Nonetheless, their work is restricted to kernel products. Finally, (1) also contains as a particular case the kde on $\mathbb{S}^{d_1} \times \mathbb{S}^{d_2}$ from García-Portugués, Crujeiras, and González-Manteiga (2015, p. 1210), where product kernels are also assumed.

3. Asymptotic Properties

To derive the asymptotic properties of estimator (1), let us first introduce $\tilde{f}$, the radial extension of f from $\mathbb{S}^d$ to $\mathbb{R}^{\tilde{d}+r} \setminus \{\mathbf{0}\}$ defined by $\tilde{f}(\mathbf{x}) := f(\mathbf{x}_1/\|\mathbf{x}_1\|, \dots, \mathbf{x}_r/\|\mathbf{x}_r\|)$. This extension allows taking derivatives of the density function f (defined only on $\mathbb{S}^d$) in a straightforward manner, with the addition of vanishing properties in those derivatives that can be explicitly controlled. The following assumptions are required:

- A1. $\tilde{f}$ is twice continuously differentiable on $\mathbb{S}^d$.
- A2. The kernel $L : \mathbb{R}_{\geq 0}^r \rightarrow \mathbb{R}_{\geq 0}$ is such that $0 < \int_{\mathbb{R}_+^r} L(\mathbf{s}) s_j^i \prod_{k=1}^r s_k^{d_k/2-1} d\mathbf{s} < \infty$ for $\ell = 1, 2, i = 0, 1$, and $j = 1, \dots, r$.
- A3. $\mathbf{h} \equiv \mathbf{h}_n$ is a sequence in $\mathbb{R}_+^r$ such that $\mathbf{h}_n \rightarrow \mathbf{0}$ and $n\rho(\mathbf{h}_n) \rightarrow \infty$ as $n \rightarrow \infty$.

Assumption A1 is needed to enable operative Taylor expansions on f via its radial extension. Note that the radial extension implies that $\nabla \tilde{f}(\mathbf{x})\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{S}^d$, that is, that the directional derivatives of $\tilde{f}$ normal to $\mathbf{x}$ are null. Assumption A2 is required to ensure the existence of first- and second-order moments of the kernel L . The condition implies that, in particular, $L(\mathbf{s}) = O(s_j^{-(d_j/2+1+\varepsilon)})$ as $s_j \rightarrow \infty$, for $\varepsilon > 0$ and $j = 1, \dots, r$. This is trivially satisfied if L has compact support, such as the Epa product kernel $L_{\text{Epa}}^P(\mathbf{s}) := \prod_{j=1}^r L_{\text{Epa}}(s_j)$, or if the kernel decays exponentially, such as the vMF product kernel L_{vMF}^P . Assumption A3 is standard to ensure the consistency of a kde.

The following theorem provides the asymptotic bias and variance expansions of (1).

Theorem 1 (Asymptotic bias and variance). Under **A1–A3**, for $\mathbf{x} \in \mathbb{S}^d$,

$$\mathbb{E}[\hat{f}(\mathbf{x}; \mathbf{h}_n)] = f(\mathbf{x}) + (\mathbf{h}_n^{\odot 2})' \mathcal{T}\bar{f}(\mathbf{x}) \mathbf{b}_d(L) + o((\mathbf{h}_n^{\odot 2})' \mathbf{1}_r), \quad (4)$$

$$\text{Var}[\hat{f}(\mathbf{x}; \mathbf{h}_n)] = \frac{v_d(L)}{n\rho(\mathbf{h}_n)} f(\mathbf{x}) + o((n\rho(\mathbf{h}_n))^{-1}), \quad (5)$$

where

$$\begin{aligned} \mathcal{T}\bar{f}(\mathbf{x}) &:= \text{diag}(\text{tr}[\mathcal{H}_{11}\bar{f}(\mathbf{x})], \dots, \text{tr}[\mathcal{H}_{rr}\bar{f}(\mathbf{x})]), \\ \mathbf{b}_d(L) &:= (b_{d,1}(L), \dots, b_{d,r}(L))', \\ b_{d,j}(L) &:= \frac{\lambda_d(L(s)s_j)}{d_j \lambda_d(L)}, \quad v_d(L) := \frac{\lambda_d(L^2)}{\lambda_d(L)^2}, \end{aligned} \quad (6)$$

where $\lambda_d(L(s)s_j) \equiv \omega_{d-1} \int_{\mathbb{R}_+^r} L(s) s_j \prod_{j=1}^r s_j^{d_j/2-1} 2^{d_j/2-1} ds$, $\mathcal{H}_{jj}\bar{f}(\mathbf{x})$ is the $(d_j + 1) \times (d_j + 1)$ matrix corresponding to the j th diagonal block of the Hessian matrix of $\bar{f}$ at $\mathbf{x}$, $\mathbf{1}_r$ is the r -dimensional vector of ones, and tr is the trace operator.

A lower order for the remaining term of the bias given in (4) can be obtained with the following stronger differentiability assumption, given that the third-order term vanishes due to symmetry.

A1*. $\bar{f}$ is four times continuously differentiable on $\mathbb{S}^d$.

Corollary 1 (Improved asymptotic bias expansion). Under **A1*** and **A2–A3**, for $\mathbf{x} \in \mathbb{S}^d$,

$$\mathbb{E}[\hat{f}(\mathbf{x}; \mathbf{h}_n)] = f(\mathbf{x}) + (\mathbf{h}_n^{\odot 2})' \mathcal{T}\bar{f}(\mathbf{x}) \mathbf{b}_d(L) + O((\mathbf{h}_n^{\odot 4})' \mathbf{1}_r).$$

Remark 1. In the case of a product kernel $L^P(\mathbf{s}) := \prod_{j=1}^r L(s_j)$, common dimensions $d := d_1 = \dots = d_r$, and equal bandwidths $h_n := h_{1,n} = \dots = h_{r,n}$, **Theorem 1** becomes

$$\begin{aligned} \mathbb{E}[\hat{f}(\mathbf{x}; h_n \mathbf{1}_r)] &= f(\mathbf{x}) + h_n^2 b_d(L) \text{tr}[\mathcal{T}\bar{f}(\mathbf{x})] + o(h_n^2), \\ \text{Var}[\hat{f}(\mathbf{x}; h_n \mathbf{1}_r)] &= \frac{v_d(L)}{nh_n^{dr}} f(\mathbf{x}) + o((nh_n^{dr})^{-1}), \end{aligned}$$

with $b_d(L) = \lambda_d(L(s)s)/(d\lambda_d(L))$, $v_d(L) = \lambda_d(L^2)/\lambda_d(L)^2$, and $\lambda_d(L) = 2^{d/2-1} \omega_{d-1} \int_{\mathbb{R}_+^r} L(s) s^{d/2-1} ds$.

An immediate consequence of **Theorem 1** is the strong consistency of $\hat{f}(\mathbf{x}; \mathbf{h}_n)$.

Corollary 2 (Strong pointwise consistency). Under **A1–A3**, for $\mathbf{x} \in \mathbb{S}^d$, $\hat{f}(\mathbf{x}; \mathbf{h}_n) \xrightarrow{\text{a.s.}} f(\mathbf{x})$ when $n \rightarrow \infty$.

The following corollary yields the Asymptotic Mean Integrated Squared Error (AMISE) optimal bandwidth and the optimal AMISE, particularized to the setting of **Remark 1**.

Corollary 3 (AMISE optimal bandwidth). Under **A1–A3**,

$$\begin{aligned} \text{AMISE}[\hat{f}(\cdot; \mathbf{h}_n)] &= \int_{\mathbb{S}^d} ((\mathbf{h}_n^{\odot 2})' \mathcal{T}\bar{f}(\mathbf{x}) \mathbf{b}_d(L))^2 \sigma_d(d\mathbf{x}) \\ &\quad + \frac{v_d(L)}{n\rho(\mathbf{h}_n)}, \end{aligned} \quad (7)$$

and, therefore,

$$\mathbf{h}_{\text{AMISE}} := \arg \min_{\mathbf{h} > 0} \text{AMISE}[\hat{f}(\cdot; \mathbf{h})]. \quad (8)$$

When the kernel is L^P , the common dimension is d , and the equal bandwidth is h_n ,

$$\text{AMISE}[\hat{f}(\cdot; h_n \mathbf{1}_r)] = h_n^4 b_d^2(L) R(\nabla^2 \bar{f}) + \frac{v_d(L)}{nh_n^{dr}}, \quad (9)$$

where $R(\nabla^2 \bar{f}) := \int_{(\mathbb{S}^d)^r} (\nabla^2 \bar{f}(\mathbf{x}))^2 \sigma_d(d\mathbf{x})$. In this case,

$$\begin{aligned} h_{\text{AMISE}} &:= \arg \min_{h > 0} \text{AMISE}[\hat{f}(\cdot; h \mathbf{1}_r)] \\ &= \left[\frac{v_d(L)}{b_d^2(L)} \times \frac{dr}{4R(\nabla^2 \bar{f})n} \right]^{1/(dr+4)} \end{aligned} \quad (10)$$

and

$$\begin{aligned} \text{AMISE}[\hat{f}(\cdot; h_{\text{AMISE}} \mathbf{1}_r)] \\ = c_{d,r} C_{d,r}(L^P) R(\nabla^2 \bar{f})^{dr/(dr+4)} n^{-4/(dr+4)}, \end{aligned} \quad (11)$$

with $c_{d,r} := (dr/4)^{4/(dr+1)} (1 + 4/(dr))$ and $C_{d,r}(L^P) := [v_d^{4r}(L) b_d^{2dr}(L)]^{1/(dr+4)}$.

We conclude the section stating the pointwise asymptotic distribution of estimator (1). To do so, we require an extra condition on the kernel that is slightly stronger than **A3**. This assumption is satisfied by L_{VMF}^P and L_{Epa}^P .

A4. For some $\delta > 0$, $\int_{\mathbb{R}_+^r} L^{2+\delta}(\mathbf{s}) \prod_{j=1}^r s_j^{d_j/2-1} ds < \infty$ for all $d_j \geq 1, j = 1, \dots, r$.

Theorem 2 (Pointwise asymptotic normality). Under **A1–A4**, for $\mathbf{x} \in \mathbb{S}^d$,

$$\sqrt{n\rho(\mathbf{h}_n)} (\hat{f}(\mathbf{x}; \mathbf{h}_n) - \mathbb{E}[\hat{f}(\mathbf{x}; \mathbf{h}_n)]) \rightsquigarrow \mathcal{N}(0, v_d(L)f(\mathbf{x})). \quad (12)$$

If, in addition, $\sqrt{n\rho(\mathbf{h}_n)} (\mathbf{h}_n^{\odot 2})' \mathbf{1}_r = O(1)$, then

$$\begin{aligned} \sqrt{n\rho(\mathbf{h}_n)} (\hat{f}(\mathbf{x}; \mathbf{h}_n) - f(\mathbf{x}) - (\mathbf{h}_n^{\odot 2})' \mathcal{T}\bar{f}(\mathbf{x}) \mathbf{b}_d(L)) \\ \rightsquigarrow \mathcal{N}(0, v_d(L)f(\mathbf{x})). \end{aligned} \quad (13)$$

Remark 2. Within the setting of equal bandwidths, condition $\sqrt{n\rho(\mathbf{h}_n)} \mathbf{h}_n^{\odot 2'} \mathbf{1}_r = O(1)$ in **Theorem 2** reads $nh_n^{\tilde{d}+4} = O(1)$. Together with **A3**, this implies that the order of the sequences of allowed bandwidths for (13) is such that $n^{-1/\tilde{d}} < h_n \leq n^{-1/(\tilde{d}+4)}$, that is, the bandwidth order needs to prioritize the reduction of bias, with the AMISE-optimal order $n^{-1/(\tilde{d}+4)}$ being a limiting case (we use $a_n < b_n$ and $a_n \leq b_n$ to denote $a_n = o(b_n)$ and $a_n = O(b_n)$, respectively).

Remark 3. Under **A1*** instead of **A1**, and still within the setting of equal bandwidths, **Corollary 1** transforms $nh_n^{\tilde{d}+4} = O(1)$ into $nh_n^{\tilde{d}+4} O(h_n^4) = o(1)$ and the sequence of allowed bandwidths is expanded to $n^{-1/\tilde{d}} < h_n < n^{-1/(\tilde{d}+8)}$.

Numerical experiments that illustrate and empirically validate **Theorems 1–2** are provided in Section B.1 of the SM.

4. Kernel Theory

4.1. Non-vMF Kernels on the Sphere

In Section B.2 of the SM the Epa kernel is seen to have a higher efficiency than the vMF kernel on $\mathbb{S}^d$, $d \geq 1$. For this reason, we provide next the explicit formulae for its normalizing constant and an effective means of simulating from it. Both tasks are well-established for the vMF kernel given its association with the vMF distribution, but have been elusive for the Epa kernel.

Proposition 1 (Normalizing constant of the Epa kernel). Let $d \geq 1$ and $h > 0$. Then,

$$c_{d,L_{\text{Epa}}}(h)^{-1} = \omega_d(1 - h^{-2}) [1 - F_d(m_h)] + \frac{\omega_{d-1} (1 - m_h^2)^{d/2}}{dh^2},$$

where $m_h := \max(-1, 1 - h^2)$, and F_d is the cumulative distribution function (cdf) of $\mathbf{y}'\mathbf{X}$ for $\mathbf{X} \sim \text{Unif}(\mathbb{S}^d)$ and an arbitrary $\mathbf{y} \in \mathbb{S}^d$. If $h \geq \sqrt{2}$, then $c_{d,L_{\text{Epa}}}(h)^{-1} = \omega_d(1 - h^{-2})$.

For $d = 1, 2$ and $h > 0$,

$$c_{1,L_{\text{Epa}}}(h)^{-1} = 2h^{-2}[(h^2 - 1) \arccos(m_h) + (1 - m_h^2)^{1/2}] \quad \text{and} \\ c_{2,L_{\text{Epa}}}(h)^{-1} = \pi(1 - m_h)(2 - h^{-2}(1 - m_h)).$$

The cdf F_d is tractable for any $d \geq 1$. Explicit formulae are available for $d = 1, 2$, while recurrence relations allow evaluating F_d for any $d > 2$ (see, e.g., sec. 2.2 in García-Portugués, Navarro-Esteban, and Cuesta-Albertos 2023).

Simulation from the pdf $L_{h,\text{Epa}}(\cdot, \boldsymbol{\mu}) := c_{d,L_{\text{Epa}}}(h)L_{\text{Epa}}((1 - \boldsymbol{\mu}'\cdot)/h^2) : \mathbb{S}^d \rightarrow \mathbb{R}_{\geq 0}$, $\boldsymbol{\mu} \in \mathbb{S}^d$, can be achieved, as in any rotationally symmetric distribution (see Wood 1994, for the vMF case), through the tangent-normal decomposition: if $\mathbf{Y} := T\boldsymbol{\mu} + \sqrt{1 - T^2}\mathbf{B}_\mu\boldsymbol{\Xi}$, where $\boldsymbol{\Xi} \sim \text{Unif}(\mathbb{S}^{d-1})$ is independent from the angular random variable $T \sim F_{d,\text{Epa},h}$ on $[-1, 1]$ and $\mathbf{B}_\mu$ is a semiorthogonal $(d+1) \times d$ matrix such that $\boldsymbol{\mu}'\mathbf{B}_\mu = \mathbf{0}$, then $\mathbf{Y} \sim L_{h,\text{Epa}}(\cdot, \boldsymbol{\mu})$.

Proposition 2 (Angular cdf of the Epa kernel). Let $d \geq 1$, $h > 0$, and $\mathbf{Y} \sim L_{h,\text{Epa}}(\cdot, \boldsymbol{\mu})$. Then, the pdf of $T = \mathbf{Y}'\boldsymbol{\mu}$ is $f_{d,\text{Epa},h}(t) = \omega_{d-1}c_{d,L_{\text{Epa}}}(h)\{1 - (1 - t)/h^2\}(1 - t^2)^{d/2-1}1_{\{m_h < t < 1\}}$. Its cdf is

$$F_{d,\text{Epa},h}(t) = \frac{\omega_{d-1}c_{d,L_{\text{Epa}}}(h)}{h^2} \times \left\{ (h^2 - 1) \frac{\omega_d}{\omega_{d-1}} [F_d(t) - F_d(m_h)] - \frac{1}{d} \left[(1 - t^2)^{d/2} - (1 - m_h^2)^{d/2} \right] \right\},$$

with $t \in [m_h, 1]$. In addition, when $d = 2$:

$$F_{2,\text{Epa},h}^{-1}(u) = 1 - h^2 \pm \sqrt{(h^2 - 1)^2 - m_h + (m_h + u(1 - m_h))(2h^2 - 1 + m_h)}, \\ u \in (0, 1).$$

We next consider the “softplus” function $\text{sfp}(t) := \log(1 + e^t)$ to build a smooth approximator of the Epa kernel that combines the higher efficiency of the latter kernel with the smoothness and positiveness of the vMF kernel. Indeed, the kernel $L_{\text{sfp}_v}(t) :=$

$\text{sfp}(v(1 - t))/\text{sfp}(v)$ for $v > 0$ is such that $\lim_{v \rightarrow \infty} L_{\text{sfp}_v}(t) = L_{\text{Epa}}(t)$ (see Figure 1(a)). The normalizing constant of L_{sfp_v} and the angular cdf $F_{d,\text{sfp}_v,h}$ are challenging for general $d \geq 1$, but they admit tractable forms for $d = 2$.

Proposition 3 (Normalizing constant and angular cdf of the sfp kernel). Let $d = 2$, $h > 0$, $v > 0$, and $\mathbf{Y} \sim L_{h,\text{sfp}_v}(\cdot, \boldsymbol{\mu})$. Then, the pdf of $T = \mathbf{Y}'\boldsymbol{\mu}$ is $f_{d,\text{sfp}_v,h}(t) = \omega_{d-1}c_{d,L_{\text{sfp}_v}}(h)L_{\text{sfp}_v}((1 - t)/h^2)(1 - t^2)^{d/2-1}$,

$$c_{2,L_{\text{sfp}_v}}(h)^{-1} = \frac{2\pi}{\text{sfp}(v)v} \left[\text{Li}_2(-e^{v(1-2h^{-2})}) - \text{Li}_2(-e^v) \right] h^2, \quad \text{and} \\ F_{2,\text{sfp}_v,h}(t) = \frac{\text{Li}_2(-e^{v(1-2h^{-2})}) - \text{Li}_2(-e^{v(1-(1-t)h^{-2})})}{\text{Li}_2(-e^{v(1-2h^{-2})}) - \text{Li}_2(-e^v)},$$

for $t \in [-1, 1]$, where Li_s is the polylogarithm function with integral representation $\text{Li}_s(z) = \Gamma(s)^{-1} \int_0^\infty t^{s-1}/(e^t/z - 1) dt$ for $s > 0$ and $z < 1$.

The following result collects the moments of the presented kernels. These appear in the results of Section 3 regarding product kernels and are key in establishing kernel efficiency in Section 4.3.

Proposition 4 (Moments of the vMF, Epa, and sfp kernels). Let $d \geq 1$. Then,

$$b_d(L_{\text{vMF}}) = \frac{1}{2}, \quad v_d(L_{\text{vMF}}) = \frac{1}{(2\pi^{1/2})^d}, \\ b_d(L_{\text{Epa}}) = \frac{1}{d+4}, \quad v_d(L_{\text{Epa}}) = \frac{4\Gamma(d/2+2)}{(2\pi)^{d/2}(d+4)}, \\ b_d(L_{\text{sfp}_v}) = \frac{\text{Li}_{d/2+2}(-e^v)}{2v \text{Li}_{d/2+1}(-e^v)}, \\ v_d(L_{\text{sfp}_v}) = \frac{v^d J_d(v)}{(2\pi)^{d/2} \Gamma(d/2) \text{Li}_{d/2+1}^2(-e^v)},$$

where $J_d(v) := \int_0^\infty \log(1 + e^{v(1-t)})^2 t^{d/2-1} dt$.

Remark 4. For all $d \geq 1$ and $h > 0$, $\lim_{v \rightarrow \infty} c_{2,L_{\text{sfp}_v}}(h) = c_{2,L_{\text{Epa}}}(h)$. Similarly, $\lim_{v \rightarrow \infty} b_d(L_{\text{sfp}_v}) = b_d(L_{\text{Epa}})$ and $\lim_{v \rightarrow \infty} v_d(L_{\text{sfp}_v}) = v_d(L_{\text{Epa}})$. These statements follow from the asymptotic equivalence $\text{Li}_s(\pm e^w) \asymp -w^s/\Gamma(s+1)$ as $w \rightarrow \infty$, for $s \in \mathbb{N}$ (Wood 1992, eq. (11.3)).

Simulation from $f_{d,\text{Epa},h}$, for general $d \geq 1$, can also be approached by acceptance-rejection: sample $T \sim f_{d,\text{vMF},h}$ and accept it if $U < L_{\text{Epa}}((1 - T)/h^2)/[ML_{\text{vMF}}((1 - T)/h^2)]$, where $U \sim \text{Unif}[0, 1]$ and $M = c_{d,L_{\text{Epa}}}(h)/c_{d,\text{vMF}}(h)$ since $L_{\text{vMF}} \geq L_{\text{Epa}}$. An analogous approach holds for $f_{d,\text{sfp}_v,h}$, with $M = c_{d,\text{sfp}_v}(h)/c_{d,\text{vMF}}(h)$ for sufficiently large $v > 0$.

4.2. Spherically Symmetric Kernels on the Polysphere

The product kernel $L^P(\mathbf{s}) = \prod_{j=1}^r L(s_j)$ from Remark 1 is an example of a kernel on $\mathbb{S}^d$ induced by a kernel $L : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ on $\mathbb{S}^d$. Another instance of such construction is the *spherically symmetric kernel* $L^S(\mathbf{s}) := L(\|\mathbf{s}\|_1)$. In the Euclidean setting, disparities are commonly observed between product and spherically symmetric kernels (Wand and Jones 1994, sec. 4.5), with

the Gaussian kernel being a notable exception. Analogously, the vMF kernel verifies that $L_{\text{vMF}}^P = L_{\text{vMF}}^S$. From the Epa and sfp kernels on $\mathbb{S}^d$, we define their spherically symmetric counterparts on $\mathbb{S}^d$ $L_{\text{Epa}}^S(\mathbf{s}) := L_{\text{Epa}}(\|\mathbf{s}\|_1) = (1 - \|\mathbf{s}\|_1)1_{\{\|\mathbf{s}\|_1 \leq 1\}}$ and $L_{\text{sfp}_v}^S(\mathbf{s}) := L_{\text{sfp}_v}(\|\mathbf{s}\|_1) = \text{sfp}(\nu(1 - \|\mathbf{s}\|_1))/\text{sfp}(\nu)$, respectively.

First, we establish a useful direct connection between the moments of any spherically symmetric kernel L^S on $\mathbb{S}^d$ and those for L on $\mathbb{S}^{\tilde{d}}$, where $\tilde{d} = \sum_{j=1}^r d_j$.

Proposition 5 (Moments of spherically symmetric kernels). If $d_1, \dots, d_r \geq 1$ and $L^S(\mathbf{s}) = L(\|\mathbf{s}\|_1)$, then $\lambda_d(L^S) = \lambda_{\tilde{d}}(L)$, $\mathbf{b}_d(L^S) = \mathbf{b}_{\tilde{d}}(L)\mathbf{1}_r$, and $\nu_d(L^S) = \nu_{\tilde{d}}(L)$.

Remark 5. Using Proposition 5, we can readily derive analogous results to (9)–(11) when the kernel is L^S , the common dimension is d , and the equal bandwidth is h_n . In this case, (9)–(10) hold replacing $b_d^2(L)$ and $\nu_d^r(L)$ with $b_{dr}^2(L)$ and $\nu_{dr}(L)$, while (11) becomes

$$\begin{aligned} \text{AMISE}[\hat{f}(\cdot; h_{\text{AMISE}}\mathbf{1}_r)] \\ = c_{d,r}C_{d,r}(L^S)R(\nabla^2 \hat{f})^{dr/(dr+4)}n^{-4/(dr+4)}, \end{aligned} \quad (14)$$

with $C_{d,r}(L^S) := [v_{dr}^A(L)b_{dr}^{2dr}(L)]^{1/(dr+4)}$.

General asymptotic formulae for the normalizing constants of L_{Epa}^S and $L_{\text{sfp}_v}^S$ are given next. For the specific $(\mathbb{S}^2)^r$ case, exact expressions are derivable.

Proposition 6 (Normalizing constants of the Epa^S and sfp_v^S kernels). For $d_1, \dots, d_r \geq 1$, $\nu > 0$, $\mathbf{h} \in \mathbb{R}_+^r$, and as $\mathbf{h} \rightarrow \mathbf{0}$,

$$\begin{aligned} c_{d,L_{\text{Epa}}^S}(\mathbf{h})^{-1} &\asymp \rho(\mathbf{h}) \frac{4(2\pi)^{\tilde{d}/2}}{\tilde{d}(\tilde{d}+2)\Gamma(\tilde{d}/2)}, \\ c_{d,L_{\text{sfp}_v}^S}(\mathbf{h})^{-1} &\asymp -\rho(\mathbf{h}) \left(\frac{2\pi}{\nu}\right)^{\tilde{d}/2} \frac{\text{Li}_{\tilde{d}/2+1}(-e^\nu)}{\text{sfp}(\nu)}. \end{aligned} \quad (15)$$

Under common dimension $d = 2$ and equal bandwidth h , then

$$\begin{aligned} c_{2,L_{\text{Epa}}^S}(h\mathbf{1}_r)^{-1} &= \frac{(2\pi h^2)^r (-1)^r}{(r+1)!} \\ &\times \sum_{\ell=\lceil r-h^2/2 \rceil}^r \binom{r}{\ell} (-1)^\ell (1 - 2h^{-2}(r-\ell))^{r+1}, \\ c_{2,L_{\text{sfp}_v}^S}(h\mathbf{1}_r)^{-1} &= \frac{(2\pi h^2)^r (-1)^{r-1}}{\text{sfp}(\nu)} \frac{1}{\nu^r} \\ &\times \sum_{\ell=0}^r \binom{r}{\ell} (-1)^\ell \text{Li}_{r+1}(-e^{\nu(1-rh^{-2})+\nu h^{-2}(2\ell-r)}). \end{aligned}$$

If $h < \sqrt{2}$, then $c_{2,L_{\text{Epa}}^S}(h\mathbf{1}_r)^{-1} = (2\pi h^2)^r/(r+1)!$.

Similarly to Section 4.1, simulation from the pdfs $L_{\mathbf{h},\text{Epa}}^S(\cdot, \boldsymbol{\mu}) := c_{d,L_{\text{Epa}}^S}(\mathbf{h})L_{\text{Epa}}(\sum_{j=1}^r (1 - \mu'_{j \cdot j})/h_j^2) : \mathbb{S}^d \rightarrow \mathbb{R}_{\geq 0}$ and $L_{\mathbf{h},\text{sfp}_v}^S(\cdot, \boldsymbol{\mu})$, $\boldsymbol{\mu} \in \mathbb{S}^d$, can be performed through acceptance-rejection from $L_{\mathbf{h},\text{vMF}}^P(\cdot, \boldsymbol{\mu})$, which admits straightforward sampling.

4.3. Kernel Efficiency

We restrict to the case of common dimension d and equal bandwidth h throughout this subsection. This simplification is motivated by $\mathbb{S}^d$ being the most practically relevant polyspherical space, and on it the components of $\mathbf{h}_{\text{AMISE}}$ are of equal order. Given that the kernel functionals $C_{d,r}(L^P)$ and $C_{d,r}(L^S)$ determine the optimal AMISE for a product kernel L^P and a spherically symmetric kernel L^S (see (11) and (14), respectively), we define the *efficiency* of the product or spherically symmetric kernel L relative to L_{Epa}^S as

$$\text{eff}_{d,r}(L) := \left[\frac{C_{d,r}(L_{\text{Epa}}^S)}{C_{d,r}(L)} \right]^{(dr+4)/4}.$$

The rationale of this definition is standard in kernel smoothing. In view of (11) and (14), $\text{eff}_{d,r}(L)$ represents the fraction of the sample that L_{Epa}^S requires to achieve the minimum AMISE of L when both are used with their AMISE bandwidths. The kernel Epa^S is taken as the reference since, as Table 1 in the SM corroborates, it is the most efficient among the explored kernels.

Proposition 7 (Efficiencies of the vMF^S, sfp_v^S, sfp_v^P, and Epa^P kernels). Let $r, d \geq 1$ and $\nu > 0$. Then:

$$\text{eff}_{d,r}(L_{\text{vMF}}^S) = \frac{2^{dr+2}\Gamma(dr/2+2)}{(dr+4)^{dr/2+1}}, \quad (16)$$

$$\begin{aligned} \text{eff}_{d,r}(L_{\text{sfp}_v}^S) &= \frac{2^{dr/2+2}\Gamma(dr/2+2)\Gamma(dr/2)|\text{Li}|_{dr/2+1}^{dr/2+2}(-e^\nu)}{(dr+4)^{dr/2+1}\nu^{dr/2}J_{dr}(\nu)|\text{Li}|_{dr/2+2}^{dr/2}(-e^\nu)}, \\ \text{eff}_{d,r}(L_{\text{sfp}_v}^P) &= \text{eff}_{d,r}(L_{\text{sfp}_v}^S) \end{aligned} \quad (17)$$

$$\begin{aligned} &\times \frac{\Gamma(d/2)^r J_{dr}(\nu)|\text{Li}|_{dr/2+2}^{dr/2}(-e^\nu)|\text{Li}|_{d/2+1}^{r(d/2+2)}(-e^\nu)}{\Gamma(dr/2)J_d^r(\nu)|\text{Li}|_{dr/2+1}^{dr/2+2}(-e^\nu)|\text{Li}|_{d/2+2}^{dr/2}(-e^\nu)}, \\ \text{eff}_{d,r}(L_{\text{Epa}}^P) &= \frac{4^{1-r}\Gamma(dr/2+2)(d+4)^{r(d/2+1)}}{\Gamma(d/2+2)^r(dr+4)^{dr/2+1}}, \end{aligned} \quad (18)$$

where $J_d(\nu)$ was defined in Proposition 4 and $|\text{Li}|_s$ denotes the absolute value of the Li_s function.

Several comments on these efficiencies follow. First, the efficiency of L_{vMF}^S with respect to L_{Epa}^S on $(\mathbb{S}^d)^r$ is exactly that of the Gaussian kernel K_{Gauss}^S with respect to the spherically symmetric Epa kernel on $\mathbb{R}^p$ with $p = dr$. Indeed, from Theorem 2 of Duong (2015), it follows that

$$\begin{aligned} \text{eff}_p(K_{\text{Gauss}}^S) &= \frac{[4\Gamma(p/2+2)/(\pi^{p/2}(p+4))](p+4)^{-p/2}}{(4\pi)^{-p/2}} \\ &= \frac{2^{p+2}\Gamma(p/2+2)}{(p+4)^{p/2+1}}, \end{aligned}$$

which coincides with (16). Second, (18) also coincides with the efficiency of the kernel $(K_{\text{Epa},d}^S)^r$ on $(\mathbb{R}^d)^r$ formed by the r -product of spherically symmetric Epa kernels on $\mathbb{R}^d$. This efficiency is also obtainable from Theorem 2 of Duong (2015):

$$\text{eff}_p((K_{\text{Epa},d}^S)^r) = \frac{[4\Gamma(dr/2+2)/(\pi^{dr/2}(dr+4))](dr+4)^{-dr/2}}{[4\Gamma(d/2+2)/(\pi^{d/2}(d+4))]^r(d+4)^{-dr/2}}$$

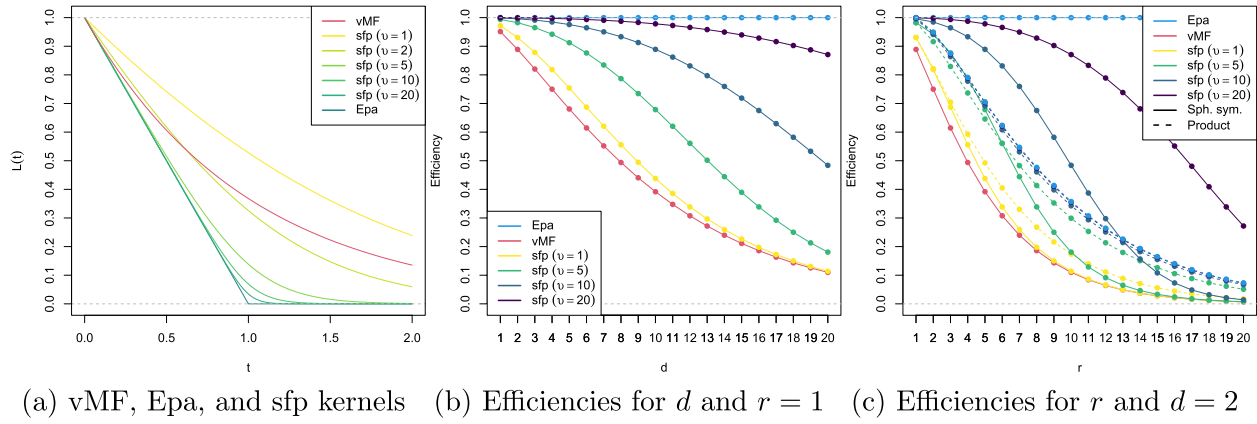


Figure 1. Kernels and their efficiencies. (a) shows the approximation of the Epa kernel by the sfp kernel as v increases. (b) displays the efficiencies of the vMF and sfp kernels with respect to the Epa^s kernel for a fixed $r = 1$. (c) does the same, but now for a fixed $d = 2$.

$$= \frac{4^{1-r} \Gamma(dr/2 + 2)(d + 4)^{r(d/2+1)}}{\Gamma(d/2 + 2)^r (dr + 4)^{dr/2+1}}.$$

Finally, the effect of (d, r) is symmetric for spherically symmetric kernels. Adding dimensions on the spherical or product space is equivalent in terms of efficiency for these kernels.

The efficiencies of the vMF, sfp, and Epa kernels are plotted in Figures 1(b) and (c), for fixed $r = 1$ and $d = 2$, respectively. Numerical results are listed in Section B.2 of the SM.

5. Bandwidth Selection

5.1. Plug-in Bandwidths

Using the properties of the PvMF density given in (3), it is possible to compute the curvature term in the AMISE optimal bandwidth for this pdf, and then derive a (zero-stage) plug-in bandwidth selector.

Proposition 8 (PvMF blockwise Hessian traces). For $\mathbf{x}, \boldsymbol{\mu} \in \mathbb{S}^d$ and $\boldsymbol{\kappa} \in \mathbb{R}_{\geq 0}^r$,

$$\mathcal{T}_{\tilde{f}_{\text{PvMF}}}(\mathbf{x}) = f_{\text{PvMF}}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\kappa}) \times \text{diag}(\psi_1(\mathbf{x}'_1 \boldsymbol{\mu}_1, \kappa_1), \dots, \psi_r(\mathbf{x}'_r \boldsymbol{\mu}_r, \kappa_r)),$$

with $\psi_j(t, \kappa) := \kappa[\kappa(1 - t^2) - d_j t]$.

Proposition 9 (AMISE optimal bandwidth for PvMF). Assume that f is a $f_{\text{PvMF}}(\cdot; \boldsymbol{\mu}, \boldsymbol{\kappa})$. Then, under **A1–A3**,

$$\text{AMISE}[\hat{f}(\cdot; \mathbf{h}_n)] = (\mathbf{h}_n^{\odot 2} \odot \mathbf{b}_d(L))' \mathbf{R}(\boldsymbol{\kappa}) (\mathbf{h}_n^{\odot 2} \odot \mathbf{b}_d(L)) + \frac{v_d(L)}{n\rho(\mathbf{h}_n)},$$

where $\mathbf{R}(\boldsymbol{\kappa})$ is the $r \times r$ matrix of curvature terms given by

$$\mathbf{R}(\boldsymbol{\kappa}) = \frac{1}{4} \left\{ \frac{1}{2} \text{diag}(\mathbf{v}(\boldsymbol{\kappa})) + (\mathbf{u}(\boldsymbol{\kappa}) \mathbf{u}(\boldsymbol{\kappa})')^{\circ} \right\} \times \prod_{j=1}^r R_0(\kappa_j),$$

$$R_0(\kappa_j) = \frac{\kappa^{(d_j-1)/2} \mathcal{I}_{(d_j-1)/2}(2\kappa_j)}{2^{d_j} \pi^{(d_j+1)/2} \mathcal{I}_{(d_j-1)/2}(\kappa_j)^2},$$

being $\mathbf{v}(\boldsymbol{\kappa}) = \mathbf{d} \odot \boldsymbol{\kappa} \odot [2(2 + \mathbf{d}) \odot \boldsymbol{\kappa} - (\mathbf{d}^{\odot 2} - \mathbf{d} + 2) \odot \mathbf{I}]$ and $\mathbf{u}(\boldsymbol{\kappa}) = \mathbf{d} \odot \boldsymbol{\kappa} \odot \mathbf{I}$, with $\mathbf{I} = (\mathcal{I}_{(d_1+1)/2}(2\kappa_1)/\mathcal{I}_{(d_1-1)/2}(2\kappa_1), \dots,$

$\mathcal{I}_{(d_r+1)/2}(2\kappa_r)/\mathcal{I}_{(d_r-1)/2}(2\kappa_r))'$, and $\mathbf{A}^{\circ}$ standing for a matrix $\mathbf{A}$ whose diagonal is set to zero. Therefore, $\mathbf{h}_{\text{AMISE}}$ is the solution to the system

$$4\mathbf{R}(\boldsymbol{\kappa}) (\mathbf{h}_{\text{AMISE}}^{\odot 2} \odot \mathbf{b}_d(L)) \odot \mathbf{h}_{\text{AMISE}} \odot \mathbf{b}_d(L) - \frac{v_d(L)}{n\rho(\mathbf{h}_{\text{AMISE}})} (\mathbf{d} \odot \mathbf{h}_{\text{AMISE}}^{\odot (-1)}) = \mathbf{0}_r.$$

For the special case of equal bandwidth h_n ,

$$h_{\text{AMISE}} = \left[\frac{\tilde{d} v_d(L)}{4\mathbf{b}_d(L)' \mathbf{R}(\boldsymbol{\kappa}) \mathbf{b}_d(L) n} \right]^{1/(\tilde{d}+4)}.$$

If, in addition, the common dimension is d , then the curvature simplifies to $R(\nabla^2 \tilde{f}_{\text{PvMF}}(\cdot; \boldsymbol{\mu}, \boldsymbol{\kappa})) = \mathbf{1}'_r \mathbf{R}(\boldsymbol{\kappa}) \mathbf{1}_r$.

If $\hat{\boldsymbol{\kappa}}$ is a suitable estimator for $\boldsymbol{\kappa}$, the Rule-Of-Thumb (ROT) bandwidth selector $\hat{\mathbf{h}}_{\text{ROT}}$ for the kde (1) with a general kernel L is the solution to the system

$$4\mathbf{R}(\hat{\boldsymbol{\kappa}}) (\hat{\mathbf{h}}_{\text{ROT}}^{\odot 2} \odot \mathbf{b}_d(L)) \odot \hat{\mathbf{h}}_{\text{ROT}} \odot \mathbf{b}_d(L) - \frac{v_d(L)}{n\rho(\hat{\mathbf{h}}_{\text{ROT}})} (\mathbf{d} \odot \hat{\mathbf{h}}_{\text{ROT}}^{\odot (-1)}) = \mathbf{0}_r. \quad (19)$$

Remark 6. If $r = 1$, the bandwidth selector $\hat{\mathbf{h}}_{\text{ROT}}$ reduces to that proposed by García-Portugués (2013, Proposition 2), taking into account that $R(\nabla^2 \tilde{f}_{\text{PvMF}}(\cdot; \boldsymbol{\mu}, \hat{\boldsymbol{\kappa}})) = d^2 R(\Psi(f_{\text{vMF}}(\cdot; \boldsymbol{\mu}, \hat{\boldsymbol{\kappa}}), \cdot))$ (see the proof of Proposition 9 of the present article).

The next algorithm defines the steps involved in the computation of the ROT bandwidth selector.

Algorithm 1 (Rule-of-thumb bandwidth selector). Given a sample $\mathbf{X}_1, \dots, \mathbf{X}_n$ on $\mathbb{S}^d$:

- Compute the maximum likelihood estimator $\hat{\boldsymbol{\kappa}}$, where $\hat{\kappa}_j = A_d^{-1}(\|\frac{1}{n} \sum_{i=1}^n \mathbf{X}_{ij}\|)$ with $A_d(r) := \mathcal{I}_{(d+1)/2}(r)/\mathcal{I}_{(d-1)/2}(r)$.
- Determine the marginal ROT bandwidths for each $\mathbb{S}^{d_j}$, $j = 1, \dots, r$:

$$\hat{h}_{j,\text{ROT}} := \left[\frac{4\pi^{1/2} \mathcal{I}_{(d_j-1)/2}(\hat{\kappa}_j)^2}{\hat{\kappa}_j^{(d_j+1)/2} [2d_j \mathcal{I}_{(d_j+1)/2}(2\hat{\kappa}_j) + (2 + d_j) \hat{\kappa}_j \mathcal{I}_{(d_j+3)/2}(2\hat{\kappa}_j)] n} \right]^{1/(4+d_j)}.$$

- iii. Solve the nonlinear equation (19) using $(\hat{h}_{1,\text{ROT}}, \dots, \hat{h}_{r,\text{ROT}})'$ as initial values.

The ROT selector assumes that the underlying density curvature can be approximated by that of a PvMF. The selector is therefore expected to perform as well as this assumption is practically met. In particular, when the underlying distribution is centro-symmetric, the selector in Algorithm 1 is expected to be outperformed by cross-validation approaches.

5.2. Cross-Validation Bandwidths

The selection of the bandwidth can also be achieved with cross-validation techniques. Specifically, one such method is the Least Squares Cross-Validation (LSCV) technique. The procedure aims to select the bandwidth by minimizing

$$\text{LSCV}(\mathbf{h}) := \int_{\mathbb{S}^d} \hat{f}(\mathbf{x}; \mathbf{h})^2 \sigma_d(\mathbf{d}\mathbf{x}) - 2n^{-1} \sum_{i=1}^n \hat{f}_{-i}(\mathbf{X}_i; \mathbf{h}), \quad (20)$$

that is, the bandwidth-dependent terms of an unbiased estimate of $\text{ISE}[\hat{f}(\cdot; \mathbf{h}_n)]$. In (20), $\hat{f}_{-i}(\mathbf{x}; \mathbf{h}) := \frac{1}{n-1} \sum_{j=1, j \neq i}^n L_{\mathbf{h}}(\mathbf{x}, \mathbf{X}_j)$ represents the leave-one-out kde.

The following result provides a closed form of (20) when using a vMF kernel.

Proposition 10 (Explicit LSCV loss for vMF kernel). If the considered kernel in (1) is L_{vMF}^P , then

$$\begin{aligned} \text{LSCV}(\mathbf{h}) = & \frac{c_d^{\text{vMF}}(\mathbf{h}^{\odot(-2)})^2}{nc_d^{\text{vMF}}(2\mathbf{h}^{\odot(-2)})} - \frac{2c_d^{\text{vMF}}(\mathbf{h}^{\odot(-2)})}{n} \\ & \times \sum_{1 \leq i < j \leq n} \left[\frac{2}{n-1} \exp \left\{ \sum_{\ell=1}^r \mathbf{X}'_{i\ell} \mathbf{X}_{j\ell} / h_\ell^2 \right\} \right. \\ & \left. - \frac{c_d^{\text{vMF}}(\mathbf{h}^{\odot(-2)})}{nc_d^{\text{vMF}}(\mathbf{a}_{ij} \odot \mathbf{h}^{\odot(-2)})} \right], \end{aligned}$$

with $\mathbf{a}_{ij,\ell} := \|\mathbf{X}_{i\ell} + \mathbf{X}_{j\ell}\|$, $\ell = 1, \dots, r$.

An alternative cross-validation approach is Likelihood Cross-Validation (LCV). In this case, the pseudo log-likelihood function

$$\text{LCV}(\mathbf{h}) := \sum_{i=1}^n \log \hat{f}_{-i}(\mathbf{X}_i; \mathbf{h})$$

is maximized. Care is needed when computing the LCV bandwidth for compactly-supported kernels, as the LCV loss is not smooth and can be minus infinite for small bandwidths. The next result is useful when performing numerical optimization with the Epa^P kernel.

Proposition 11 (Critical LCV bandwidth for Epa^P kernel). Consider the kernel L_{Epa}^P and a common bandwidth $\mathbf{h} = h\mathbf{1}_r$. Then, $\text{LCV}(\mathbf{h}) > -\infty$ if and only if $h > h_{\min, \text{Epa}}$, where

$$h_{\min, \text{Epa}}^2 := \max_{i=1, \dots, n} \min_{j \neq i} \max_{k=1, \dots, r} \{1 - \mathbf{X}'_{ik} \mathbf{X}_{jk}\}.$$

In the non-common bandwidth case $\mathbf{h}$, $\min_{k=1, \dots, r} h_k > h_{\min, \text{Epa}}$ implies $\text{LCV}(\mathbf{h}) > -\infty$.

6. Kernel-based Homogeneity Testing

The kde (1) can be used to construct a nonparametric test of the homogeneity of two classes, which will be applied later in Section 7.2. Let f_1 and f_2 denote two pdfs on $\mathbb{S}^d$. The two sample problem is formalized as

$$\mathcal{H}_0 : f_1 = f_2 \quad \text{versus} \quad \mathcal{H}_1 : f_1 \neq f_2. \quad (21)$$

Different test statistics can be constructed for this testing problem considering distinct distances between f_1 and f_2 . The one with the longest tradition is the L_p -distance

$$d_p(f_1, f_2)^p := \int_{\mathbb{S}^d} |f_1(\mathbf{x}) - f_2(\mathbf{x})|^p \sigma_d(\mathbf{d}\mathbf{x}) \quad (22)$$

with $p = 2$; see Mammen (1992, chap. 3), Anderson, Hall, and Titterton (1994), and Li (1996) for tests based on $d_2(f_1, f_2)^2$ on $\mathbb{R}^d$ and, for example, Boente, Rodriguez, and González-Manteiga (2014) for the use of $p = 1, 2$ in the context of goodness-of-fit testing on $\mathbb{S}^d$. However, (22) is remarkably unpractical in high-dimensional settings due to the vanishing of the densities f_1 and f_2 , which is amplified by a power $p > 1$. On the high-dimensional polysphere $(\mathbb{S}^2)^{168}$, the setting in the real data application of Section 7.2, this vanishing can easily yield numerically null distances when working with machine double precision. The Hellinger distance $d_H(f_1, f_2)^2 := 1 - \int_{\mathbb{S}^d} \sqrt{f_1(\mathbf{x})f_2(\mathbf{x})} \sigma_d(\mathbf{d}\mathbf{x})$ slightly alleviates the issue of vanishing densities in (22), but it is still not enough for high-dimensional setups, where working on the logarithmic scale of the density is imperative. Furthermore, these two distances involve computing numerically an integral on $\mathbb{S}^d$, which is inexact and expensive, even if adapting exact formulae for (22) when $p = 2$ and f_1 and f_2 are mixtures of vMFs (García-Portugués, Crujeiras, and González-Manteiga 2013, Proposition 4).

We use the Jensen–Shannon Divergence (JSD) introduced by Lin (1991) to measure the closeness between the density functions f_1 and f_2 on $\mathbb{S}^d$:

$$\text{JSD}(f_1, f_2) := \pi_1 \text{D}_{\text{KL}}(f_1 \| f_0) + \pi_2 \text{D}_{\text{KL}}(f_2 \| f_0),$$

$$\text{D}_{\text{KL}}(f_j \| f_0) := \int_{\mathbb{S}^d} \log \left(\frac{f_j(\mathbf{x})}{f_0(\mathbf{x})} \right) f_j(\mathbf{x}) \sigma_d(\mathbf{d}\mathbf{x}),$$

where $f_0(\mathbf{x}) := \pi_1 f_1(\mathbf{x}) + \pi_2 f_2(\mathbf{x})$ is the pooled density for both classes, π_1 and π_2 denote the prior probability of each class, and $\text{D}_{\text{KL}}(f_j \| f_0)$ is the Kullback–Leibler divergence of f_j from f_0 . The JSD is a symmetrized version of the Kullback–Leibler divergence that satisfies $0 \leq \text{JSD}(f_1, f_2) \leq \log(2)$, with $\text{JSD}(f_1, f_2) = 0$ if and only if $f_1 = f_2$ almost everywhere (Lin 1991). Its variant $\sqrt{\text{JSD}(f_1, f_2)}$ also satisfies the triangular inequality and is thus a proper distance (Endres and Schindelin 2003). However, this transformation is unnecessary when using $\text{JSD}(f_1, f_2)$ as a test statistic.

Lin (1991) also introduced the generalized JSD to compare $k \geq 2$ distributions:

$$\begin{aligned} \text{JSD}(f_1, \dots, f_k) &:= \sum_{j=1}^k \pi_j \text{D}_{\text{KL}}(f_j \| f_0) \\ &= H(f_0) - \sum_{j=1}^k \pi_j H(f_j), \end{aligned} \quad (23)$$

$$\begin{aligned} H(f_j) &:= - \int_{\mathbb{S}^d} \log(f_j(\mathbf{x})) f_j(\mathbf{x}) \sigma_d(d\mathbf{x}) \\ &= - \mathbb{E}_{f_j} [\log(f_j(\mathbf{X}_j))], \end{aligned}$$

with $f_0(\mathbf{x}) := \sum_{j=1}^r \pi_j f_j(\mathbf{x})$ and where $H(f_j)$ is the Shannon entropy of $\mathbf{X}_j \sim f_j$. The generalized JSD satisfies $0 \leq \text{JSD}(f_1, \dots, f_k) \leq \log(k)$, with $\text{JSD}(f_1, \dots, f_k) = 0$ if and only if $f_1 = \dots = f_k$ almost everywhere. This generalized JSD also allows tackling the generalized version of (21):

$$\begin{aligned} \mathcal{H}_0 : f_1 = \dots = f_k \quad \text{versus} \\ \mathcal{H}_1 : \exists i, j \in \{1, \dots, k\} : f_i \neq f_j. \end{aligned} \quad (24)$$

Let $\{\mathbf{X}_{i,1}\}_{i=1}^{n_1}, \dots, \{\mathbf{X}_{i,k}\}_{i=1}^{n_k}$ be k independent samples, each of them respectively iid with the pdfs $f_1, \dots, f_k$. The sample sizes are such $n_1 + \dots + n_k = n$. We define a test statistic for (24) as the following kernel-based estimator of (23):

$$T_{n,\text{JSD}} := \hat{H}(\hat{f}_0(\cdot; \mathbf{h})) - \sum_{j=1}^k \hat{\pi}_j \hat{H}(\hat{f}_j(\cdot; \mathbf{h})), \quad (25)$$

where $\hat{\pi}_j := n_j/n$, $j = 1, \dots, k$, are the relative frequencies of each class, and the Shannon entropies are estimated with

$$\begin{aligned} \hat{H}(\hat{f}_j(\cdot; \mathbf{h})) &:= -\frac{1}{n_j} \sum_{i=1}^{n_j} \log(\hat{f}_j^{-i}(\mathbf{X}_{i,j}; \mathbf{h})) \quad \text{and} \\ \hat{H}(\hat{f}_0(\cdot; \mathbf{h})) &:= -\frac{1}{n} \sum_{j=1}^k \sum_{i=1}^{n_j} \log(\hat{f}_0^{-(i,j)}(\mathbf{X}_{i,j}; \mathbf{h})) \end{aligned} \quad (26)$$

for the leave-one-out cross-validated kdes

$$\begin{aligned} \hat{f}_j^{-i}(\mathbf{X}_{i,j}; \mathbf{h}) &:= \frac{1}{n_j - 1} \sum_{\substack{\ell=1 \\ \ell \neq i}}^{n_j} L_{\mathbf{h}}(\mathbf{X}_{i,j}, \mathbf{X}_{\ell,j}) \quad \text{and} \\ \hat{f}_0^{-(i,j)}(\mathbf{X}_{i,j}; \mathbf{h}) &:= \frac{1}{n-1} \left\{ \sum_{\substack{m=1 \\ m \neq j}}^k \pi_m n_m \hat{f}_m(\mathbf{X}_{i,j}; \mathbf{h}) + \pi_j (n_j - 1) \hat{f}_j^{-i}(\mathbf{X}_{i,j}; \mathbf{h}) \right\}. \end{aligned} \quad (27)$$

Several comments on the construction of (25) are in order. First, the expectations in the Shannon entropies are estimated by their empirical versions. This approach is much more stable than carrying out a Monte Carlo approximation of $H(f_j)$ with a simulated sample from $\hat{f}_j(\cdot; \mathbf{h})$, especially since in high dimensions the kde is sparse and can spike at the sample observations, hence, bringing variability to the final estimate. Second, the terms $L_{\mathbf{h}}(\mathbf{X}_{i,j}, \mathbf{X}_{i,j})$ are excluded to remove unnecessary bias and avoid numerical overflows. Third, the same bandwidths are used for the k kdes $\hat{f}_j(\cdot; \mathbf{h})$, $j = 1, \dots, k$, to balance their contributions in the test statistic and to avoid different smoothing biases, this being a common practice (see, e.g., Mammen 1992; Li 1996).

Our test for (24) based on the test statistic $T_{n,\text{JSD}}$ rejects $\mathcal{H}_0$ for large values of this statistic. Theorem 3 shows that this test is consistent against all (fixed) alternatives $\mathcal{H}_1$ where $f_1 = \dots = f_k$ does not hold almost everywhere: $\mathbb{P}_{\mathcal{H}_1}[r_n(T_{n,\text{JSD}} - \mu_n) > c_\alpha] \rightarrow 1$ as $n \rightarrow \infty$, where the centering sequence μ_n , the positive sequence $r_n \rightarrow \infty$, and c_α are such that

$\mathbb{P}_{\mathcal{H}_0}[r_n(T_{n,\text{JSD}} - \mu_n) > c_\alpha] \rightarrow \alpha$ as $n \rightarrow \infty$. The result requires some additional assumptions to ensure that f_j and $\hat{f}_j(\cdot; \mathbf{h})$ are bounded away from zero, and to control the decay rate of the latter:

A5. f is bounded away from zero.

A6. The kernel L is a positive function decreasing in each of its arguments.

A7. For some fixed $C, a, b > 0$, $L(\mathbf{s}) \geq Ce^{-a\|\mathbf{s}\|_1^{2b}}$ as $\mathbf{s} \rightarrow \infty$, and $n\rho(\mathbf{h}_n)h_{n,j}^{2b} \rightarrow \infty$ as $n \rightarrow \infty$ for each $j = 1, \dots, r$.

Remark 7. The first part of **A7** is satisfied for L_{vMF} , $L_{\text{sf}_v}^P$, and $L_{\text{sf}_v}^S$ with $b \geq 1$. It is not satisfied for L_{Epa}^P and L_{Epa}^S (which are ruled out anyway by **A6**). For $b = 1$ and common bandwidths, the bandwidth condition in **A7** becomes $nh^{\tilde{d}+2} \rightarrow \infty$, that is, strengthening **A3**'s $h_n > n^{-1/\tilde{d}}$ into $h_n > n^{-1/(\tilde{d}+2)}$, which is still satisfied by the AMISE-optimal bandwidth of order $n^{-1/(\tilde{d}+4)}$ from Corollary 3.

Theorem 3. For $j = 1, \dots, k$, let $\{\mathbf{X}_{i,j}\}_{i=1}^{n_j}$ be an iid sample of the pdf f_j satisfying **A1** and **A5**. Assume that $\hat{\pi}_j = n_j/n \rightarrow \pi_j > 0$ as $n \rightarrow \infty$, with $n = n_1 + \dots + n_k$. Under **A2–A4**, **A6**, and **A7**, $T_{n,\text{JSD}} \xrightarrow{\mathbb{P}} \text{JSD}(f_1, \dots, f_k)$ as $n \rightarrow \infty$.

The p -value of the test can be easily approximated with permutations, by shuffling the class labels under $\mathcal{H}_0$. For each $b = 1, \dots, B$, a permutation resample of $T_{n,\text{JSD}}$ is

$$T_{n,\text{JSD}}^{*b} := \hat{H}^{*b}(\hat{f}_0^{*b}(\cdot; \mathbf{h})) - \sum_{j=1}^k \hat{\pi}_j \hat{H}^{*b}(\hat{f}_j^{*b}(\cdot; \mathbf{h})),$$

where $\hat{H}^{*b}(\hat{f}_0^{*b}(\cdot; \mathbf{h}))$ and $\hat{H}^{*b}(\hat{f}_j^{*b}(\cdot; \mathbf{h}))$ are defined as in (26)–(27), but with the sample $\{\mathbf{X}_{i,j}\}_{i=1}^{n_j}$ replaced with the permuted sample $\{\mathbf{X}_{i,j}^{*b}\}_{i=1}^{n_j}$ that is randomly extracted without replacement from $\{\mathbf{X}_{i,j}\}_{i=1}^{n_j}$. The p -value of the permutation test is approximated as $B^{-1} \sum_{b=1}^B \mathbf{1}_{\{T_{n,\text{JSD}}^{*b} > T_{n,\text{JSD}}\}}$.

The proposed test is nonparametric, and thus its main advantage is the absence of strong distributional assumptions. With the aim of critically assessing the performance of this test in Section B.3 of the SM, we next construct two possible competing parametric tests. These tests evaluate the equality of population location and scatter measures for the two-sample problem. For the random vectors $\mathbf{X}_j = (\mathbf{X}'_{1,j}, \dots, \mathbf{X}'_{r,j})'$, $j = 1, 2$, on the sphere $\mathbb{S}^d$, define the location vectors $\boldsymbol{\mu}_j := \mathbb{E}[\mathbf{X}_j]/\|\mathbb{E}[\mathbf{X}_j]\|$ and scatter matrices $\boldsymbol{\Sigma}_j := \mathbb{E}[\mathbf{X}_j \mathbf{X}_j']$, their empirical counterparts $\hat{\boldsymbol{\mu}}_j := \bar{\mathbf{X}}_j/\|\bar{\mathbf{X}}_j\|$ and $\hat{\boldsymbol{\Sigma}}_j := n_j^{-1} \sum_{i=1}^{n_j} \mathbf{X}_{i,j} \mathbf{X}_{i,j}'$, and the associated testing problems $\mathcal{H}_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $\mathcal{H}_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ and $\mathcal{H}_0 : \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$ versus $\mathcal{H}_1 : \boldsymbol{\Sigma}_1 \neq \boldsymbol{\Sigma}_2$. For each spherical component $\mathbb{S}^{d_\ell}$ in $\mathbb{S}^d$, $\ell = 1, \dots, r$, consider the marginal versions of the above estimators: $\hat{\boldsymbol{\mu}}_{\ell,j}$ and $\hat{\boldsymbol{\Sigma}}_{\ell,j}$. We define the location and scatter test statistics as the maximum discrepancy between the two marginal estimators along the r spheres:

$$T_{n,\text{loc}} := \max_{\ell=1, \dots, r} \|\hat{\boldsymbol{\mu}}_{\ell,1} - \hat{\boldsymbol{\mu}}_{\ell,2}\|_2,$$

$$T_{n,\text{sc}} := \max_{\ell=1, \dots, r} d_{\mathcal{P}(d_\ell+1)}(\hat{\boldsymbol{\Sigma}}_{\ell,1}, \hat{\boldsymbol{\Sigma}}_{\ell,2}),$$

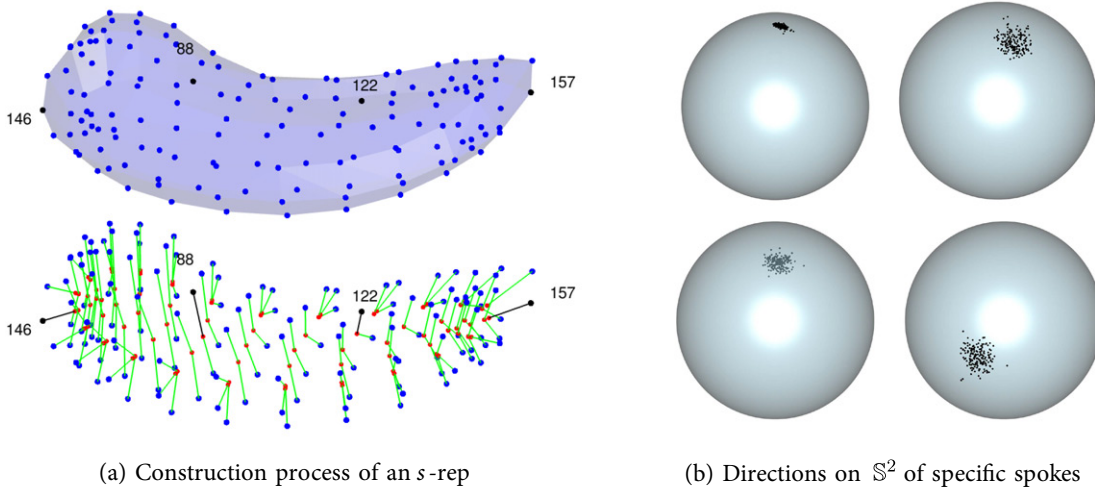


Figure 2. Construction of an s -rep and scatterplots of directions of different spokes. (a) sketches the s -rep fitting process (Liu et al. 2021), where a hippocampus surface (blue, top) is parameterized as a set of base skeletal points (red) and a collection of 168 spokes (green segments) connecting with the boundary (blue points). (b) shows the $\mathbb{S}^2$ -directions of the $n = 177$ spokes located at boundary points 88, 122, 146, and 157 (from left to right and top to bottom). The locations of these spokes are shown in (a).

where $d_{\mathcal{P}(d_\ell+1)}(\Sigma_{\ell,1}, \Sigma_{\ell,2}) = \|\log(\lambda_\ell)\|_2$, with $\lambda_\ell = (\lambda_{1,\ell}, \dots, \lambda_{(d+1),\ell})'$ the eigenvalues of $\Sigma_{\ell,1}^{-1}\Sigma_{\ell,2}$, is the affine-invariant/Fisher–Rao metric in the space $\mathcal{P}(d_\ell + 1)$ of positive definite matrices of size $d_\ell + 1$ (Pennec, Fillard, and Ayache 2006). The tests based on $T_{n,\text{loc}}$ and $T_{n,\text{sc}}$ can be calibrated by permutations.

Simulations showing the satisfactory performance of the k -sample test can be found in Section B.3 of the SM.

7. Real Data Application

We analyze the hippocampi dataset briefly presented in Section 1. The data comprises shapes of hippocampi from 177 6-months-old infants, which have been parametrized as s -reps (Liu et al. 2021). Figure 2(a) depicts the process of constructing the s -rep, where the hippocampus surface is parameterized as a collection of inner skeletal points (red) and a set of 168 spokes (green), which are vectors originating from the skeleton and extending to the boundary points (blue).

The analyzed dataset reveals variations in the shapes of the hippocampus, but the internal skeletal structures exhibit somewhat similar configurations. As a result, it is justifiable to use the average internal skeletal configuration as a common reference point, and analyze the vectors that emanate from this reference point and extend to the boundary. To focus specifically on the shape of the hippocampus, we standardize the vectors by normalizing their radii to match the sample means. This standardization removes size information and results in a simplified representation observed within the polysphere $(\mathbb{S}^2)^{168}$, capturing the hippocampus shape as an s -rep. Figure 2(b) shows the $\mathbb{S}^2$ -directions of the $n = 177$ subjects at four boundary points.

The space $(\mathbb{S}^2)^{168}$ is high-dimensional, at least according to smoothing standards. Consequently, there are many computational challenges involved in evaluating (1) and (25). The first and most pressing issue is the density vanishing, resulting in numerical underflows. We countermeasured it by (i) evaluating the log-kde and (ii) computing the density with respect to the

uniform measure $1/\omega_d$ (this is only beneficial if $\omega_d > 1$, which for a common dimension d happens if $d \leq 17$). Second, numerical stability through safe computations is also important. Applying (i) involves a recurrent use of the “LogSumExp trick” to safely evaluate the logarithms of sums. The normalizing constants for the sfp_v^S and Epa^S kernels can be approximated with the asymptotic forms (15) (without altering the density rankings or the decision of the permutation test). For $d = 2$, a numerically-stable form for the normalizing constant of the vMF kernel is $\log(c_{2,L_{\text{vMF}}}(h)) = -[\log(2\pi) + 2\log(h) + \log 1p(-\exp\{-2/h^2\})]$. Finally, we addressed the slowness of repeatedly evaluating (1) by (i) implementing in C++ the kde and (ii) designing computationally-tractable statistics such as $\hat{h}_{\text{ROT}}$ and $T_{n,\text{JSD}}$.

7.1. Inward–Outward Ranking of Hippocampi

The first application of the kde (1) involves establishing a non-parametric inward–outward ranking of the hippocampi based on the density of their shapes. Denoting by $\mathbb{X} = \{X_1, \dots, X_n\} \subset (\mathbb{S}^2)^{168}$ the $n = 177$ hippocampus shapes, we establish the rank of each subject X_i within the sample through the ranking function

$$\begin{aligned} \text{rank}(X_i; \mathbb{X}) &:= \text{rank}(\ell_{(i)}(\mathbf{h}); \ell_1(\mathbf{h}), \dots, \ell_n(\mathbf{h})), \\ \ell_i(\mathbf{h}) &:= \log \hat{f}^{-i}(X_i; \mathbf{h}), \end{aligned} \quad (28)$$

where $\text{rank}(x; x_1, \dots, x_n)$ is the standard ranking function of the univariate sample $\{x_1, \dots, x_n\}$. In (28), the logarithm does not affect the ranking, but it is fundamental for numerical stability. Similarly, removing the contribution of the i th point in $\hat{f}^{-i}(X_i; \mathbf{h})$ does not alter the ranking, but has the advantage of excluding the easily-overflowing addend $c_{d,L}(\mathbf{h})L(\mathbf{0})$ from within the logarithm.

The ranking (28) was implemented with the sfp_v^S kernel with $v = 100$ due to its higher efficiency than the vMF kernel and positivity. The bandwidth vector was selected using $\hat{h}_{\text{ROT}}$ as described in Algorithm 1. This choice is justified by the

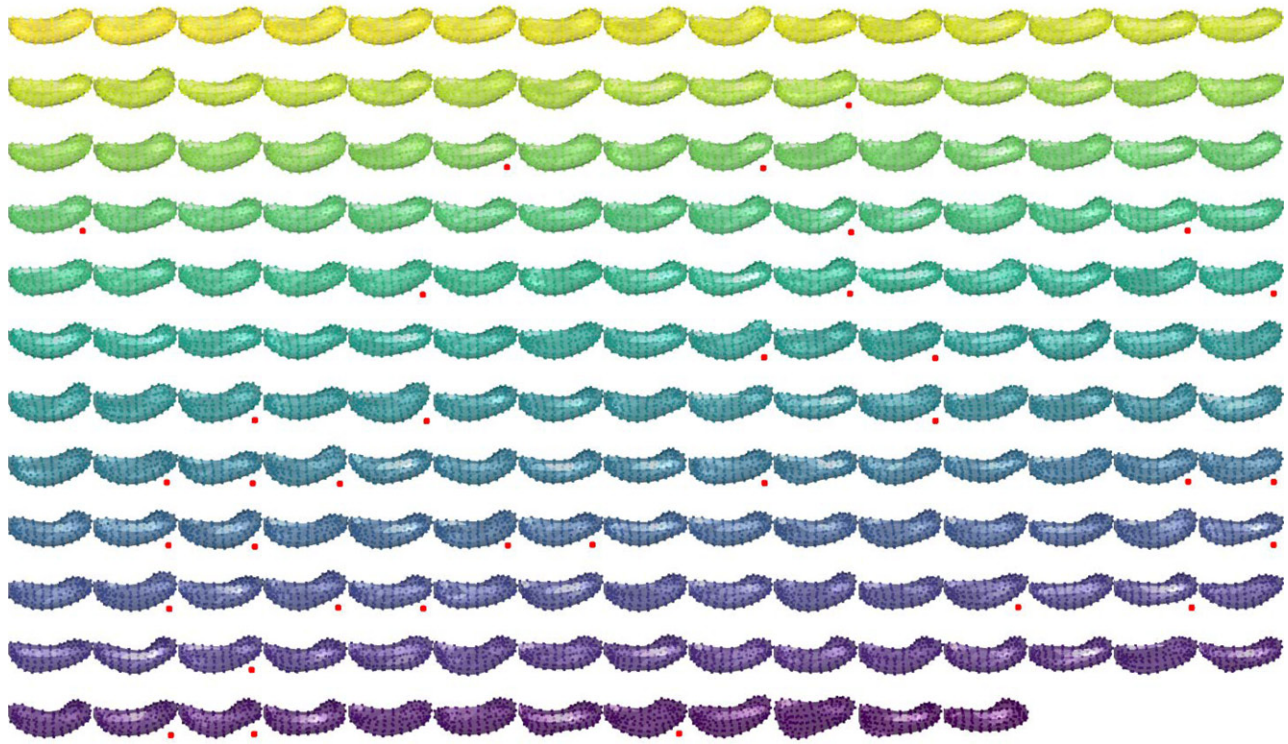


Figure 3. Dataset of $n = 177$ hippocampi ranked by the density of their shapes, as determined by (28) using $\hat{h}_{\text{ROT}}$ and the sfp kernel with $\nu = 100$. The yellow–violet color gradient indicates the inward–outward ranking. The red dots indicate whether the 6-month-old infant later developed autism (see Section 7.2).

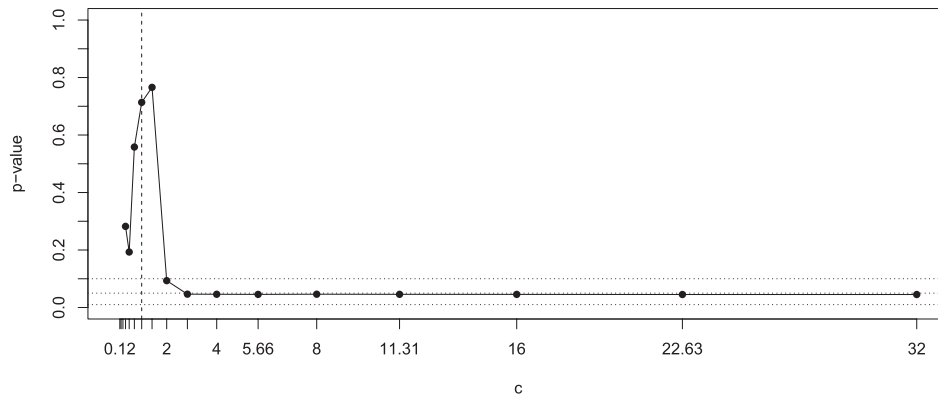


Figure 4. p -values of the test based on $T_{n,\text{JSD}}$ for bandwidth $c \times \hat{h}_{\text{ROT}}$, as a function of the factor c .

unimodality of the distributions of the spokes directions (see Figure 2(b)).

The resulting ranking sorts the hippocampi shapes in terms of their density within the sample: the first subjects represent prototypical hippocampi in high-density regions, while the last ones represent outlying hippocampi from low-density regions. Figure 3 shows the subjects sorted according to their density rank, with a yellow–violet color gradient indicating the inward–outward ranking. The subjects shown in yellow/violet correspond to the most central/extreme ones, and can be regarded as the most prototypical/outlying, respectively.

7.2. Homogeneity of Classes

The JSD two-sample test from Section 6 represents the second application of the kde (1). This test has an attractive application

in the hippocampi dataset: determining whether there exist significant shape differences, as measured by the polyspherical parameterization of s -reps, between the autistic and non-autistic classes. A preliminary analysis on the separability of these two groups was conducted with the ranking induced in Figure 3, which revealed no distinguishable distribution pattern of the subjects in the autistic class. In particular, the autistic class does not have the most inward or outward subjects in terms of density. However, this analysis is not conclusive of the existence of significant differences between the distributions of both classes.

To test the homogeneity of the two classes, we used $B = 5000$ permutations, a range of bandwidths $c \times \hat{h}_{\text{ROT}}$, $c = 2^\ell$ with $\ell \in \{-3, -2.5, \dots, 5\}$, and the sfp kernel with $\nu = 100$. The selection of this bandwidth range is informed by the experience with E2 in Section B.3 of the SM. Figure 4 indicates a mild significant difference between the two classes, with p -values

slightly below the 5% significance level for at least $c \geq 3$ and p -values on the 10% significance level for $c = 2$. Observe that in E2 the largest power gains occurred for $c \in \{2, 4, 8, 16, 32\}$. The mild significant difference found is consistent with that reported in Liu et al. (2023b, sec. 7.4), where, among others, a two-sample test based on DiProPerm (Wei et al. 2016) and PNS scores was applied (p -value = 0.028). In summary, their test works on the PNS scores of the hippocampus preshapes. Therefore, as ours, their analysis considers a framework where the size information of the hippocampi is removed.

The location and scatter tests gave p -values 0.2524 and 0.3372, using as well $B = 5000$ permutations, thus, failing to reject. The proposed JSD test is therefore adding a clear value to this particular application over parametric approaches. These tests also fail to reject when inspecting the differences between groups in each of the 168 marginal $\mathbb{S}^2$ -distributions and correcting the resulting p -values of each test to control the False Discovery Rate (FDR; Benjamini and Yekutieli 2001). The FDR-corrected p -values, omitted, are exactly one. The rejection of the null hypothesis also does not happen for the JSD test run on each spoke, for the range of c 's considered for $(\mathbb{S}^2)^{168}$ and with FDR-corrected p -values across spokes (virtually all the FDR-corrected p -values are one). Hence, the significant differences between classes are subtle and appear in the joint variation of the spokes on $(\mathbb{S}^2)^{168}$, not in the marginal distributions on $\mathbb{S}^2$.

8. Discussion

This article introduces a kernel density estimator for data on the polysphere $\mathbb{S}^d$ and, as a spin-off, a k -sample test that is tailored to be applicable in relatively high dimensions. The mean square error analysis of the estimator provides an inexpensive bandwidth selector that is particularly useful for estimating unimodal densities. The (non-product) kernel theory allows quantifying kernel efficiencies for the first time in directional supports. The estimator, bandwidth selector, and k -sample test are shown to be practically relevant in analyzing s -reps of hippocampi: the density-based ranking serves to identify prototypical and outlying subjects, and the homogeneity test reveals mild significant differences between classes that cannot be captured by exploring the marginal variation of spokes.

The density theory in this article is focused on squared-error asymptotics and fixed dimension. Alternative developments, to be tackled elsewhere, involve studying the asymptotics for normalized density estimation with both sample size and dimensions diverging to infinity, as well as the derivation of uniform confidence regions and L^p -norm consistencies (relatedly, see Bai, Rao, and Zhao 1988; Wu and Wu 2022). More specific future research avenues are outlined next. First, there is room for improvement in bandwidth selection, either by using plug-in selectors based on mixtures (e.g., mixtures of vMFs) or bootstrap selectors, or by developing asymptotic theory for cross-validators selectors. Second, a natural extension of the kernel efficiency analysis will be the more general framework of different bandwidths and spherical dimensions. A final research endeavor is the derivation of the asymptotic distribution of the proposed Jensen–Shannon divergence test statistic and the exploration of its power under local alternatives.

Supplementary Materials

Supplement: The supplementary material include detailed proofs of the results, numerical experiments (assessing asymptotic normality, kernel efficiency, and the validity and relevance of the k -sample test), as well as additional plots related to the real data application.

Software: The R package `polykde` (available at <https://CRAN.R-project.org/package=polykde>) implements the proposed methods and enables replication of the empirical results.

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Disclosure Statement

The authors report there are no competing interests to declare.

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